OGLE/2MASS/Gaia Measurements of the Large & Small Magellanic Clouds

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ABSTRACT

The Magellanic Clouds have been studied for 30 years by the OGLE variable star survey, with archives containing the mean V & I band magnitudes and periods of thousands of Cepheids and RR Lyrae. These standard candles' absolute magnitudes are functions of period for Cepheid, and, functions of period and metallicity for RR Lyrae. Cross-matching with 2MASS for HJK bands, the apparent magnitude, distance modulus *(m-M)*, and distance are calculated for in the VIHJK bands independently and the final distance is the weighted numerical mean of each band equally for LMC Cepheids IVJK and SMC Cepheids, and VIHJK for the SMC RR Lyrae. The OGLE/2MASS photometric data with Gaia astrometric cuts yields 61.2 kpc with a dispersion of 7.2 kpc. For the LMC we utilized RR Lyrae relations calibrated to Cepheids, and the RR Lyrae and Cepheids in the SMC have 51.2 kpc with a dispersion of 6 kpc.

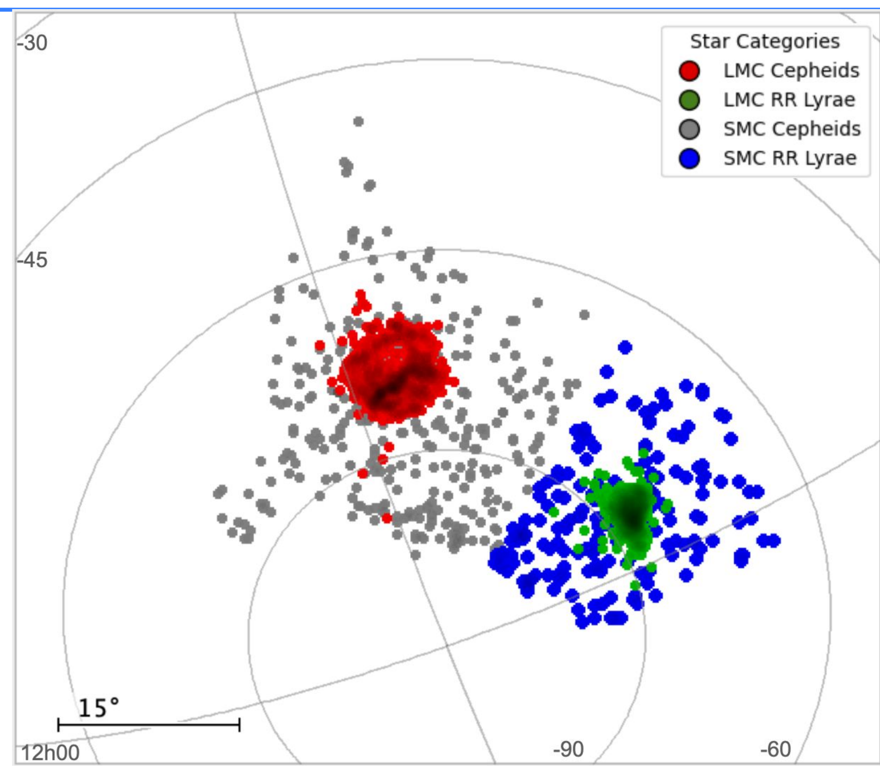

Fig 1. Initial sky plot of the OGLE Cepheids and RR Lyrae within the LMC and SMC. Spatial outliers indicate field stars prior to crossmatching 2MASS and astrometric filtering.

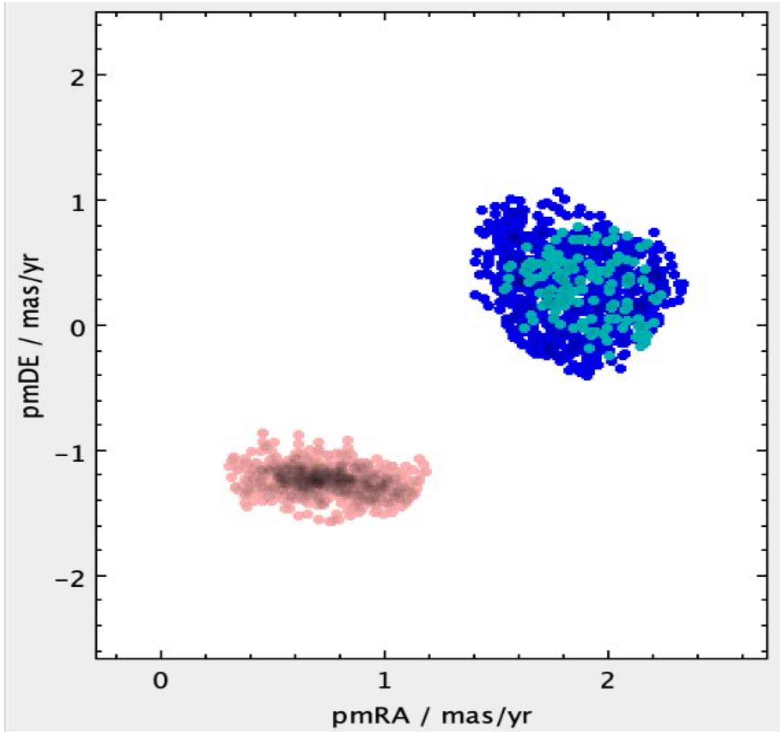

Fig 2. Proper motion plot (PMRA, PMDec) after trimming for outliers. .

The SMC clusters near (0.8, −1.2) mas/yr and the LMC near (1.8, 0.5) mas/yr.

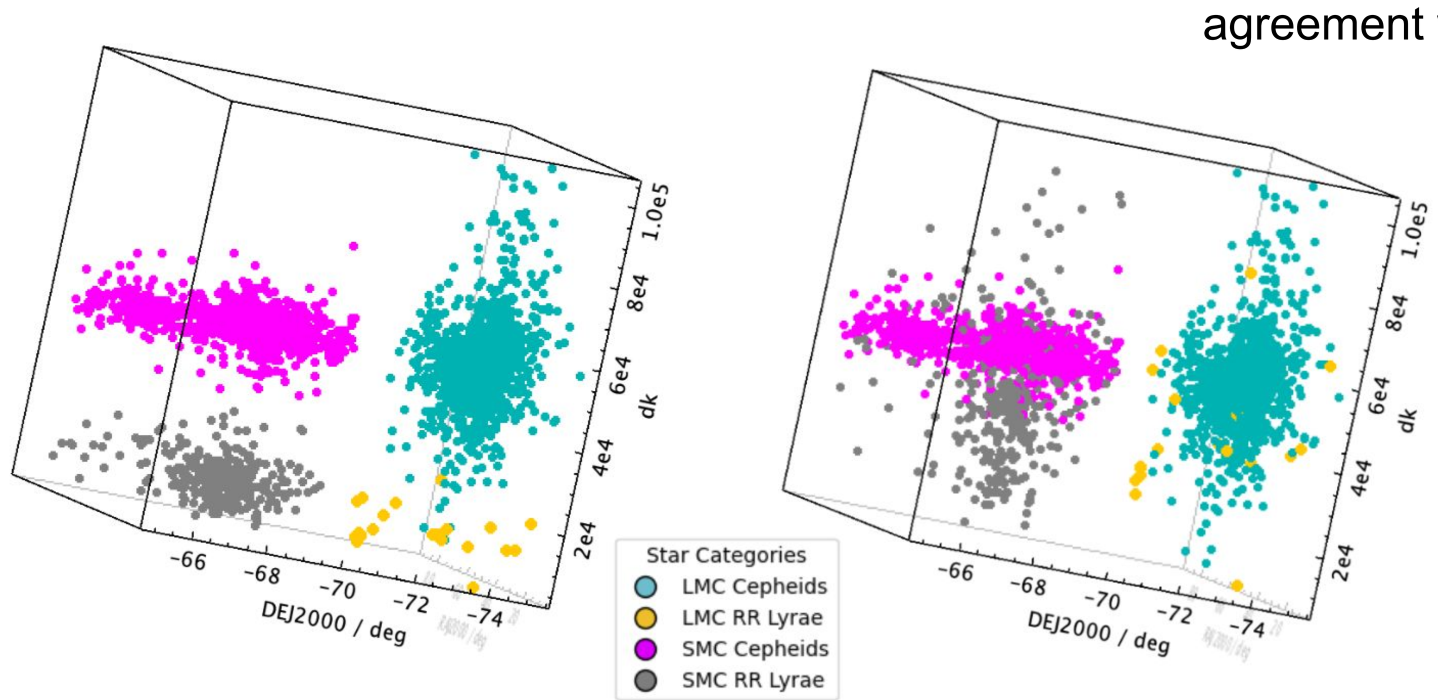

Fig 3. RA–Dec–distance projections showing the line-of-sight distance spread of Cepheids and RR Lyrae in the LMC and SMC.

Adjustments to the period luminosity relationship were conducted in order to visually shift the RR Lyrae distances to roughly match the Cepheids.

METHODS

Cross-matching with 2MASS provided H, J, and K photometry, and distances were calculated independently in the available V, I, H, J, and K bands.A cross-match with Gaia DR3 produced unrealistic negative parallaxes for these distant sources; instead, proper motion cuts in PMRA and PMDec were used to separate LMC and SMC populations. During this process, SMC RR Lyrae stars were removed, as they appeared as outliers in proper motion space.

Cepheid distances were derived using published period-luminosity relations in the V, I, J, and K bands calibrated for the Milky Way, LMC, and SMC. RR Lyrae distances were derived using period-luminosity-metallicity relations in the V, I, H, J, and K bands. Initial RR Lyrae H-band distances were systematically offset (around 17 kpc). Visual comparison with the consistent V, I, J, and K distances showed this discrepancy, and the H-band relation was shifted to bring it into agreement with the VIJK distance scale.

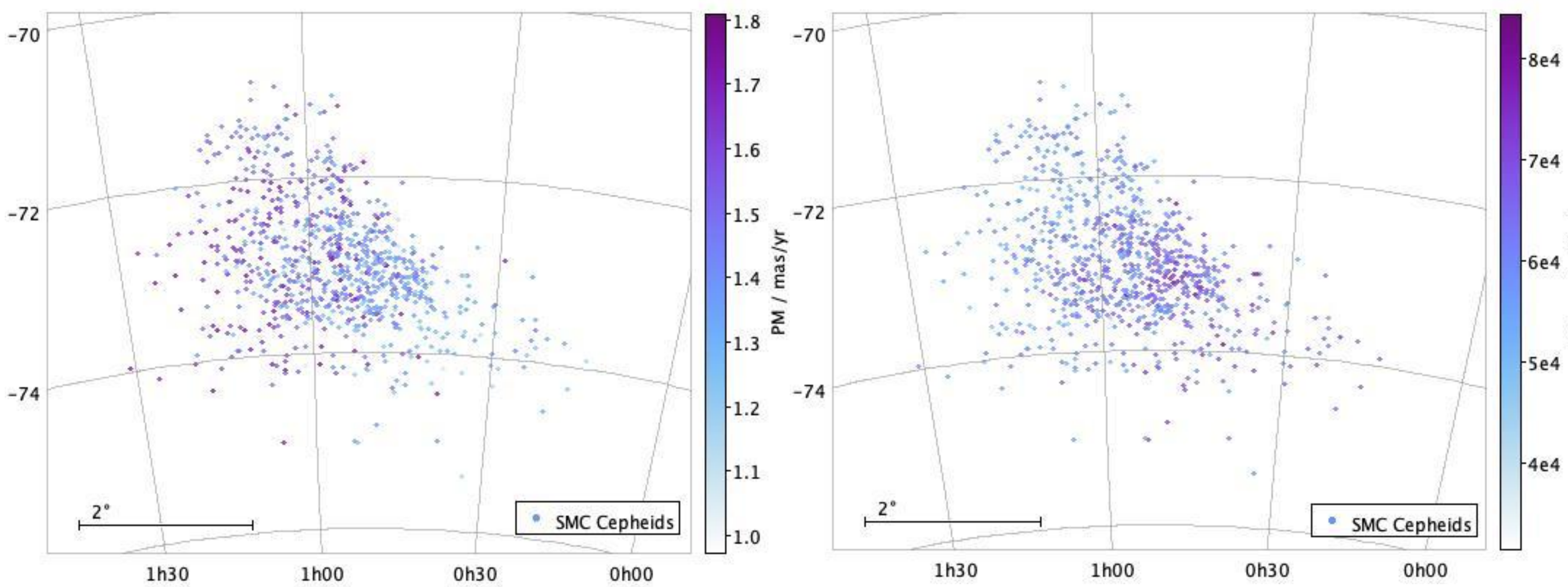

Fig 4. SMC Cepheid skyplot weighted by proper motion (1-1.8) mas/yr on the left, and distance (32-82) kpc on right. One side is systematically further and shows smaller proper motions, indicating a tilt along the line of sight.

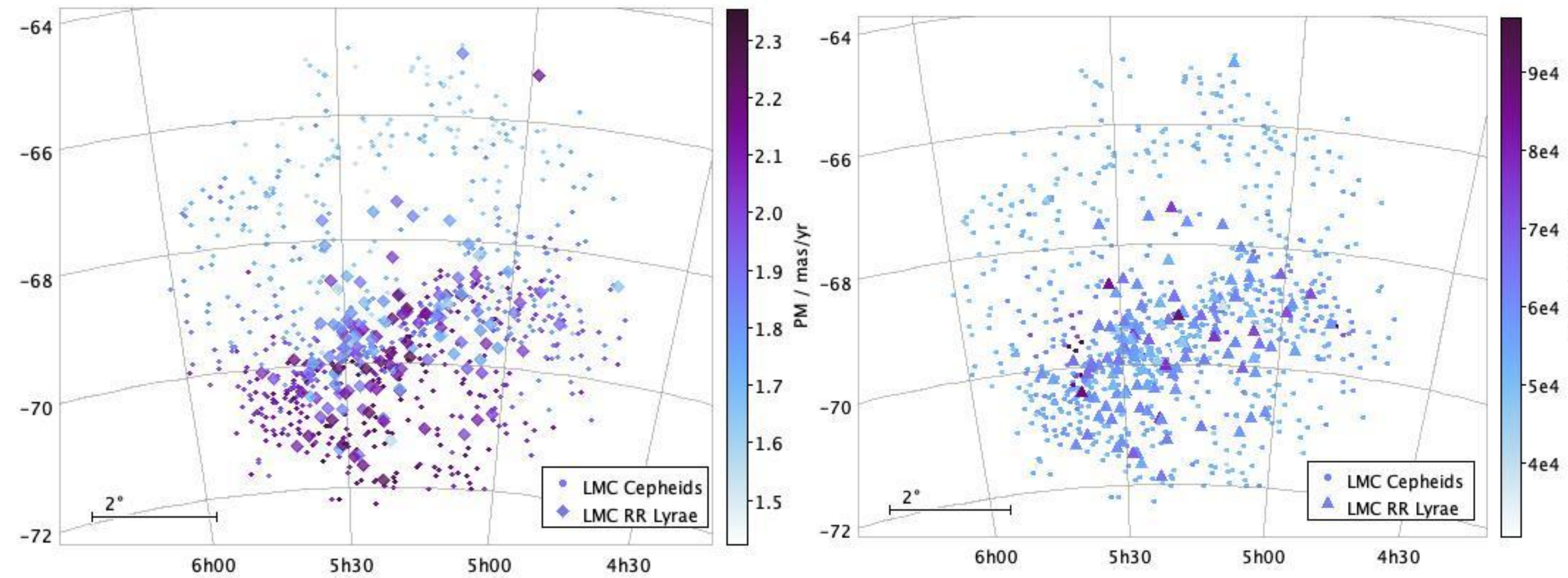

Fig 5. LMC Cepheids and RR Lyrae with proper motion (1.4 - 2.4) mas/yr mapped on the left with distance (30-100) kpc on the right. This shows trace of the bar and disk structure in the more dense regions. Unlike the SMC, no corresponding distance/PM gradient is visible across the sky.

Cepheids

Variable	Mean	STD
mu i	18.92472	0.1669
mu v	18.95235	0.209611
mu j	19.05167	0.214307
mu k	18.74823	0.358778
di	61126.06889	4694.69
dv	62014.03142	5980.73
dj	64930.85914	6433.8
dk	56961.43854	9542.8

SMC Cepheid distance: $\boxed{61.2 \pm 7.2 \text{ kpc}}$

d = 62.44 kpc (Graczyk, et. al 2020)

Cepheids

Variable	Mean	STD
mu i	18.62265	0.149951
mu v	18.74741	0.215976
mu j	18.42419	0.175375
mu k	18.44262	0.167758
di	53159.39243	3753.76
dv	56451.61753	5816.65
dj	48556.66428	3909.78
dk	48957.64921	3790.43

RR Lyrae

Variable	Mean	STD
mu i	18.60475	0.280092
mu v	18.7661	0.289996
mu J	18.05526	0.728881
mu k	17.91095	0.72292
mu h	18.07723	0.786701
di	53011.17004	6358.07
dv	57156.59765	7604.71
dj	43074.79818	13444.
dk	40276.76269	12644.
dh	43902.92989	14886.

LMC RR Lyrae & Cepheid distance: $\boxed{51.2 \pm 6.0 \text{ kpc}}$

d= 49.97 kpc (Pietrzynski et al 2013)

*Table 1. Distance moduli (**mu = m-M = 5log(d) +5**) and distances in VIJK for Cepheids within the LMC & SMC, and VIHJK for the LMC RR Lyraes.*

INTERPRETATIONS & FUTURE WORK

Mapping distance and proper motion across the sky shows that the Magellanic Clouds possess real line-of-sight structure in depth. This is strongest in the SMC, where one side of the galaxy appears systematically farther away and exhibits smaller proper motions, consistent with a tilted geometry rather than random contamination. In contrast, the LMC panels clearly trace the bar and disk structure of the galaxy but do not show a strong distance/PM gradient corresponding with a tilt, where the closer section yields a larger proper motion. Numerically, the SMC Cepheid sample yields a mean distance of 61.2 kpc with a dispersion of 7.2 kpc, while the LMC combined Cepheid and RR Lyrae sample yields a mean distance of 51.2 kpc with a dispersion of 6.0 kpc, the dispersions reflect the average spreads of these stellar distances in each band. The RR Lyrae analysis was the least stable component of the pipeline: although OGLE reports ~44,000 RR Lyrae in the LMC field, most are likely foreground or background contaminants once proper motion and photometric cuts are applied, and the 2MASS requirement is especially restrictive, retaining only ~2% of the sample. The RR Lyrae relations used here appear too averaged for this population, with initial near-infrared distances clustering near ~17 kpc and requiring visual calibration that may shift foreground stars up.

Future improvements include testing known Magallenic RR Lyrae relations, repeating the analysis using OGLE V and I photometry with no 2MASS crossmatch- Gaia proper motions only, and extending the method to additional OGLE variable classes with well-studied period-luminosity relations.

REFERENCES

The OGLE Variable Star Catalog (Udalski et al.), the Two Micron All Sky Survey (2MASS; Skrutskie et al. 2006), and Gaia DR3 astrometry (Gaia Collaboration et al. 2023) were used in this work. Adopted distance references include the Small Magellanic Cloud distance from Graczyk et al. (2020, *ApJ*, 904, 13) and the Large Magellanic Cloud distance from Graczyk et al. (2013, *ApJ*, 780, 59). Cepheid period–luminosity relations were taken from Gieren et al. (2018, *A&A*, 620, A99), who examine metallicity effects using Baade–Wesselink analyses across the Milky Way and Magellanic Clouds. RR Lyrae period–luminosity–metallicity relations were adopted from Catelan, Pritzl, & Smith (2004, *ApJS*, 154, 633).